

LOW-COST AUTOMATION

Using PIC16F676

PAMARTHI KANAKARAJA

There are many types of home and industrial automation systems available in the market. These include IR remote controller, Bluetooth home automation, DTMF home automation, Wi-Fi home automation, RF home automation, and voice controlled home automation. The one presented here is a low-cost automation system. It can be used at home or in an industrial environment to switch devices on and off.

IR remote controllers and receivers follow standard protocols for sending and receiving the data. Some of the standard protocols are NEC, PHILIPS RC5, JVC, and SIRC (Sony infra-red remote control). We will be using the NEC protocol for this project. After understanding the frame format of IR remote, we will interface IR receiver TSOP1738

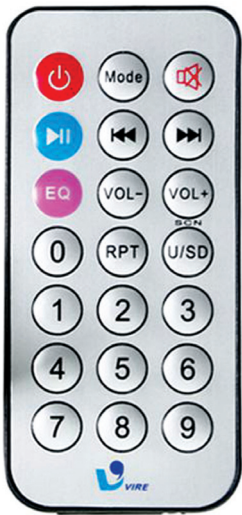


Fig. 1: VIRE NEC IR remote

with

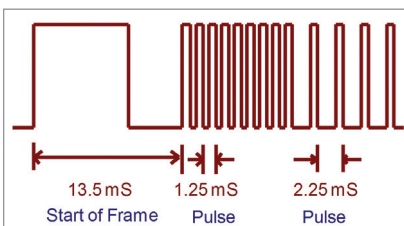


Fig. 2: Pulse distance frame

PIC16F676 microcontroller to decode the key pressed from an NEC IR remote, as shown in Fig. 1.

Each bit is transmitted using the pulse distance, as shown in Fig. 2. Logical '0' is a 562.5μs pulse burst followed by a 562.5μs space, with a total transmit time of 1.125ms. Logical '1' is a 562.5μs pulse burst followed by a 1.6875ms space, with a total transmit time of 2.25ms.

When a key is pressed on the remote controller, the message transmitted consists of the following in the order below:

1. A 9ms leading pulse burst (16 times the pulse burst length used for a logical data bit)
2. A 4.5ms space
3. The 8-bit address for the receiving device
4. The 8-bit logical inverse of the address
5. The 8-bit command
6. The 8-bit logical inverse of the command
7. A final 562.5μs pulse burst to

signify the end of message transmission

The four bytes of data bits each are sent with least significant bit first. Fig. 3 illustrates the format of an NEC IR transmission frame for an address of 00h (00000000b) and a command of ADh (10101101b).

A total of 67.5ms is required to transmit a message frame. It needs 27ms to transmit the 16 bits of address (address + inverse) and the 16 bits of command (command + inverse).

Table 1 shows 32-bit NEC frame format and Table 2 shows the complete list of IR key codes for the NEC IR remote.

PIC16F676 controller

PIC16F676 is Microchip's 8-bit CMOS PIC microcontroller with flash memory. It comes in a 14-pin package with high-performance RISC CPU, which makes it an ideal choice for most electronic applications related to embedded systems

TABLE 1
32-BIT NEC FRAME FORMAT

Address	Compliment of Address	Command	Compliment of Command
LSB-MSB (0-7)	LSB-MSB (8-15)	LSB-MSB (16-23)	LSB-MSB (24-31)

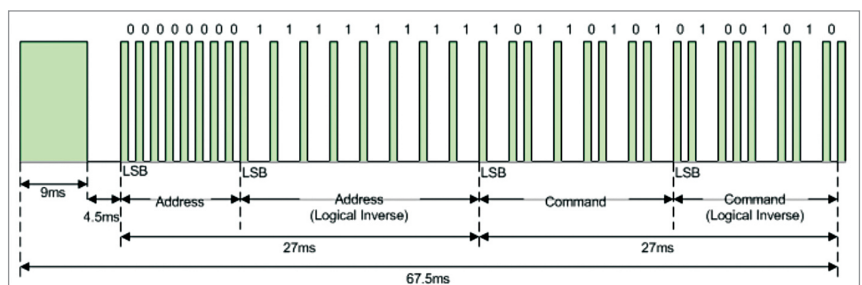


Fig. 3: Format of an NEC IR transmission frame